



INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

End-Spring Semester Examination 2022-23

Date of Examination: 20 April 2023 Session: AN Duration: 3 hrs. Full Marks: 100

Subject No.: AI61004 Subject: STATISTICAL FOUNDATIONS OF AI/ML

Department/Center/School: CENTRE OF EXCELLENCE IN ARTIFICIAL INTELLIGENCE

Specific charts, graph paper, log book etc., required NO

Special Instructions (if any): WRITE SHORT BUT PRECISE ANSWERS SHOWING ALL MATHEMATICAL DETAILS.

ANSWER ANY 5 OF THE QUESTIONS BELOW. ALL PARTS OF EACH QUESTION MUST BE ANSWERED TOGETHER

Q1: A) There is a Gaussian distribution $N(u, 9)$. Consider a prior distribution $N(0, 25)$ on 'u'. You receive data sequentially from the Gaussian distribution [5.1, 4.3 7.7 5.4 2.1 4.8]. Show how the posterior and MAP estimation of 'u' evolve as this data arrives one by one? Derive the necessary expressions.

[10 marks]

B) Prove that if Y follows chi-squared(n), then $E(Y)=n$ and $Var(Y)=2n$.

[5 marks]

C) Given an example of an estimator which is biased but asymptotically unbiased, with necessary derivation.

[5 marks]

Q2: A) Prove that the samples obtained by rejection sampling actually follow the target distribution. [5 marks]

B) You want to calculate the expectation of a Gamma Distribution $Ga(4, 3)$ using Importance Sampling using a suitable Gaussian distribution. Using the following pseudorandom number sequence in (1,100), demonstrate the process using 10 samples.

Random number sequence: [82 91 13 32 64 10 28 55 96 47]

[8 marks]

C) What are the pros and cons of the proposal distribution having high variance in case of Metropolis-Hastings?

[4 marks]

D) Explain the significance of "burn-in phase" in MCMC methods.

[3 marks]

Q3: A) You have drawn 3 samples $x_1 \sim \text{Gamma}(a, 1)$, $x_2 \sim \text{Gamma}(b, 1)$, $x_3 \sim \text{Gamma}(c, 1)$. Consider $y=(y_1, y_2, y_3)$ where $y_i = x_i / (x_1 + x_2 + x_3)$. Prove that $y \sim \text{Dirichlet}(a, b, c)$. [8 marks]

B) Discuss how we can carry out Gibbs Sampling to estimate the latent variables of a GMM. You can consider all the Gaussian component means $(\mu_1, \mu_2, \dots, \mu_K)$ as latent variables following a

Gaussian prior $N(0, s^2)$ and the proportion Π of mixture components as following a Dirichlet prior with parameters $(\alpha_1, \alpha_2, \dots, \alpha_K)$. The Gaussian mixture variances $(\sigma_1, \sigma_2 \dots \sigma_K)$ can be considered as equal and known (σ) . Show the joint distribution and the conditionals required for Gibbs Sampling. [12 marks]

Q4. A) Consider the following model: $Y_i \sim \text{Ber}(p)$, $Z_i \sim U(0,1)$, $X_i \sim N(W_k Z_i, s)$ where $k = Y_i$

We have 3 observations of X and Z : $[X_1=3.6 \ Z_1=0.2, X_2=-1.2 \ Z_2=0.5, X_3=4.2 \ Z_3=0.8]$

Consider the following 2 choices of parameters:

$P_1: [W_1=-5, W_2=5, s=9, p=0.5]$, $P_2: [W_1=0, W_2=4, s=25, p=0.8]$

Which of these two sets of parameters fits the observations better? We do not observe Y . [8 marks]

B) Show that E-M algorithm for GMMs is analogous to K-means clustering under certain parameter settings. Can you suggest any change to K-means to make it more similar to E-M? (hint: compare the E-M steps and K-means steps)

[8 marks]

C) In which circumstances can E-M lead to wrong estimates of parameters of a GMM?

[4 marks]

Q5. A) How can you compare an Auto-Regressive model and an HMM on the given sequential data?

[6 marks]

Data sequence $X = [-2.2, 0.9, 1.3, -1.8]$

AR model: $X(1) \sim N(0,4)$, $X(t+1) \sim N(1.1 * X(t), 4)$

HMM model (hidden states S_1, S_2): $\pi = [0.5 \ 0.5]$ (Initial state distribution)

$A(S_1, S_1) = A(S_2, S_2) = 0.8$ (Transition Distribution)

$X(t)|Z(t)=S_1 \sim N(-2, 1)$, $X(t)|Z(t)=S_2 \sim N(1, 1)$ (Emission Distribution)

B) You have a dataset X of 10 binary observations. Your Null Hypothesis is that the Bernoulli parameter is 0.5, while the alternate hypothesis is that this parameter is below 0.5. Your test-statistic $T(X)$ is the fraction of '1's, and your rejection region is defined by $(T(X) - 0.5 < \epsilon)$. What will be possible values of ϵ , if you are to reject the null hypothesis at 5% level of significance?

[4 marks]

C) You have two datasets: $A = [2.3 \ 3.1 \ 2.8]$ and $B = [3.0 \ 3.6 \ 5.2]$. The null hypothesis is that, they follow the same distribution. Test this hypothesis using the Permutation test over any 10 (randomly chosen) permutations, at 5% level of significance.

[5 marks]

D) A dataset of 20 observations are divided into 5 groups of 4 observations each. The maximum values of these groups are x_1, x_2, x_3, x_4, x_5 . If the data are known to follow a distribution for which the 90th percentile is 'c', what is the expected number of groups for which the group maximum value will exceed 'c'?

[5 marks]

Q6. A) Draw 10 samples $X = (x_1, x_2, \dots, x_{10})$ from $\text{Beta}(5, 10)$, using rejection sampling (you can ignore the denominators of Beta PDF for sampling). Use the pseudorandom number sequence below for this purpose. **[4 marks]**

B) Using each of the above sampled values, draw samples $Y = (y_1, y_2, \dots, y_{10})$ such that $y_i \sim N(x_i, 4)$ using the same pseudorandom number sequence. **[4 marks]**

C) Fit Kernel-based Density-based functions $f(X)$ and $g(Y)$ on these samples using Gaussian Kernels. For both X and Y , write the expressions of the fitted density functions at any arbitrary point. **[2 marks]**

D) Calculate the difference between the original and fitted distributions (both X and Y) at 3 uniformly spaced points within the range of the samples. **[4 marks]**

E) Discuss, without numerical calculations, how you can use a Copula Function $C(\theta)$ to estimate a joint density $h(X, Y)$ using the fitted functions $f(X)$ and $g(Y)$. **[4 marks]**

F) Finally, how will you evaluate the quality of the fitted joint distribution $h(X, Y)$ using a suitable test of hypothesis? **[2 marks]**

Random number sequence: [82 91 13 92 64 10 28 55 96 97 16 98 72 49 81 15 43 22
80 37 66 4 85 94 68 76 75 40 66 18]